2️⃣ TECHNICAL REQUIREMENTS DOCUMENT (TRD)

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1. System Architecture Overview

Architecture follows a three-tier model:

Frontend (Next.js + TypeScript)
↓
Backend API (Django + DRF)
↓
Database (Supabase PostgreSQL)
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Private Object Storage (Supabase Storage)

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2. Technology Stack

Frontend

- Next.js (App Router)
- TypeScript
- React Query
- Tailwind CSS
- Axios/Fetch
- Role-based route guards

Backend

- Django
- Django REST Framework
- JWT Authentication
- Custom permission classes
- UUID Case ID generator
- Severity classifier (rule-based)
- Signed URL generation logic

Database

- Supabase PostgreSQL
- Relational schema with constraints
- Encrypted at rest

Storage

- Supabase Private Bucket
- Signed URL access (time-limited)
- File validation middleware

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3. Core Database Schema

users

- id
- name
- email
- password_hash
- role (admin / hr)
- is_active
- created_at
- updated_at

complaints

- id
- case_id (UUID, unique)
- category
- title
- description
- incident_date
- location
- accused_name
- accused_role
- impact_flags
- urgency_level
- status
- resolution_note
- created_at
- updated_at

complaint_attachments

- id
- complaint_id (FK)
- file_name
- file_path
- file_type
- file_size

- uploaded_at

No reporter-related table exists.

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4. API Structure

Public

- POST /api/access-key/validate
- POST /api/complaints/
- GET /api/complaints/track/{case_id}/

HR

- POST /api/hr/login
- GET /api/hr/complaints
- PATCH /api/hr/complaints/{id}
- GET /api/hr/analytics

Admin

- POST /api/admin/login
- POST /api/admin/hr-users
- PATCH /api/admin/hr-users/{id}
- GET /api/admin/overview

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5. Security Design

- JWT tokens stored in HTTP-only cookies.
- Middleware-based role enforcement.
- Signed URL generation endpoint.
- Rate limiting for anonymous endpoints.
- Input validation & sanitization.
- CSRF protection enabled.

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6. Deployment Strategy

- Frontend deployed on Vercel / Cloud platform.
- Backend deployed on scalable container environment.
- Supabase managed PostgreSQL.
- Supabase Storage private bucket.